DHAK Activity Assay Coupled to NADH Oxidation

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

This protocol measures the activity of dihydroxyacetone kinase (DHAK), either in the cell-free extract (CFE) or after affinity purification. DHAK phosphorylates dihydroxyacetone (DHA) to dihydroxyacetone phosphate (DHAP) using ATP. DHAP has no useful absorbance, so it is reduced to glycerol-3-phosphate by α-glycerophosphate dehydrogenase (α-GDH), which oxidises one NADH to NAD^+ per DHAP; triosephosphate isomerase (TIM) is included in the coupling mix. Only NADH absorbs at 340 nm, so the initial rate of decrease in A_{340} after adding DHA is proportional to DHAK activity. The NADH extinction coefficient (ε_{340} = 6220 M^{-1} cm^{-1}) converts this rate into µmol of DHAP formed per minute, a value comparable between instruments and laboratories.

Materials list

- DHAK sample: cell-free extract (CFE, Protocol 5) or purified and concentrated DHAK (after affinity purification)
- **50 mM potassium phosphate buffer (KH_2PO_4), pH 7.5**
- **ATP** stock (**5 mM final**)
- **$MgCl_2$, 0.5 M stock** (**5 mM final**)
- **NADH** stock, freshly prepared and kept on ice in the dark (**0.2 mM final**)
- **Dihydroxyacetone (DHA)** stock (**2.5 mM final**)
- **α-GDH** and **TIM** coupling enzymes (about **1 U of each** per reaction)
- UV-Vis spectrophotometer with kinetic mode at **340 nm**
- **1 cm** path-length cuvettes (UV-transparent)
- Micropipettes and tips, ice

Steps

1. Prepare all stock solutions in 50 mM potassium phosphate buffer, pH 7.5. Keep NADH, ATP and the enzymes on ice.
2. Set the spectrophotometer to **340 nm** in kinetic mode and zero it with **1 mL of buffer** in the cuvette.
3. In a **1 mL** final volume, add to the cuvette everything except DHA: buffer, **ATP (5 mM)**, **$MgCl_2$ (5 mM)**, **NADH (0.2 mM)**, about **1 U of α-GDH** and **1 U of TIM**, and the DHAK sample (see step 4). Leave room for the DHA volume. Mix gently by pipetting.
4. Amount of DHAK. If the sample is CFE, start with **5 µL** and increase the volume in successive assays until a clear decrease in A_{340} is observed; then repeat the assay with that adjusted volume. If the sample is purified DHAK, add about **6 U**, or titrate in the same way until activity is observed.
5. Baseline. Record A_{340} for **1–2 min**. The NADH signal should read about **1.0–1.2** (0.2 mM NADH gives a theoretical A_{340} of 1.24) and stay flat.
6. Once the baseline is stable, remove the cuvette and add **DHA to 2.5 mM final** to start the reaction.

7. Mix by pipetting up and down and return the cuvette **immediately** to the spectrophotometer.
8. Record A_{340} over time until the decrease is clearly defined. Note the time between DHA addition and the first reading.
9. Determine the slope of the **initial linear region** (ΔA_{340}/min) and subtract the slope of the baseline from step 5.
10. Calculate the activity as **$U/mL = (\Delta A_{340}/min \times V_total) / (\varepsilon_{340} \times l \times V_enzyme)$**, with ε_{340} = 6.22 mM^{-1} cm^{-1}, l = 1 cm, V_total = 1 mL and V_enzyme the volume of DHAK sample added (in mL). One unit (U) is 1 µmol of DHAP formed per minute. For example, with 5 µL of sample, U/mL = ΔA_{340}/min × 32.2; with 1 µL, U/mL = ΔA_{340}/min × 160.8.
11. Divide by the protein concentration of the DHAK sample (mg/mL) to obtain the specific activity in **U/mg**.
12. Run each measurement at least in **triplicate**.

Troubleshooting

- **Baseline not flat:** NADH is being consumed without DHA, from NADH-oxidising enzymes in the sample (more likely in CFE than in purified DHAK) or from NADH degradation. Subtract the baseline slope and prepare NADH fresh.
- **Initial A_{340} far from about 1.2:** Check the NADH stock. NADH degrades in solution, especially at acidic pH and room temperature.
- **Absorbance drops too fast to measure an initial rate:** Dilute the DHAK sample and use the same volume, so the decrease stays linear for at least 1 min.
- **Coupling enzymes limiting:** α-GDH and TIM must be in excess so that DHAK sets the rate. Doubling the coupling mix should not change the measured rate.
- **No decrease after DHA addition:** Confirm that ATP and $MgCl_2$ were added, since DHAK requires Mg-ATP, and that the enzyme was not inactivated during purification.
- **Buffer choice:** The reference thesis assay uses Tris-HCl at pH 8.0. Here **50 mM potassium phosphate at pH 7.5** is used, consistent with the lysis and purification buffers. Activities are therefore not directly comparable with the reference values.